9.2 Ramsey's Theorem in Miniature

WORKSHEET

The question

STEP I 1989, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s1-q4

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself The six points matter only through which pairs are joined by which colour: the geometry is a distraction. How many segments are there, and how many leave each point?

STAGE 1 Five segments, two colours

Covers: six points

Ask yourself Five segments leave the point A, each gold or silver. Can at most two of them be each colour? What must be true of at least three of them?

STAGE 2 Look at the far ends

Covers: six points

Pause and try Say AB, AC, AD are all gold. Now look at triangle BCD. What happens if any one of its sides is gold? And if none is?

STAGE 3 Five points are not enough

Covers: five points

Ask yourself With five points each point has only four segments, so you could try two of each colour at every point. Place the points as a regular pentagon: colour the sides one colour and the diagonals the other. Is there a one-colour triangle?

What's really going on

Ramsey numbers

The *Ramsey number* $R(s,t)$ is the smallest N such that every colouring of the edges of K_N in gold and silver contains either a gold K_s (a set of s points with all edges between them gold) or a silver K_t. The question has shown that $R(3,3) = 6$: six points always work, and the pentagon shows five do not.

Frank Ramsey proved in 1928 that these numbers always exist: for every s and t there is some N that works. The proof is an induction that generalises the argument above, and shows $R(s,t) \le R(s-1,t) + R(s,t-1)$. But exact values are extraordinarily hard to find:

$$R(3,3) = 6, \qquad R(4,4) = 18, \qquad 43 \le R(5,5) \le 46.$$

The value of $R(4,4)$ was found in 1955. Despite enormous computer searches, $R(5,5)$ is still unknown; the upper bound of 46 was proved only in 2024. A computer cannot simply check every colouring of K_{43}: there are 2^{903} of them, far more than the number of atoms in the universe.

Erdős and the aliens

Paul Erdős, who did more than anyone to develop Ramsey theory, liked to tell this story. Suppose aliens invade the Earth and threaten to destroy it unless we tell them the value of $R(5,5)$. We should marshal all our computers and mathematicians, and we might find it. But if they asked for $R(6,6)$, we should try to destroy the aliens. Erdős also proved, in 1947, that $R(k,k) > 2^{k/2}$, using a startling method: he showed that a *randomly* coloured graph on fewer than $2^{k/2}$ points has a positive probability of containing no one-colour K_k, so a good colouring must exist, even though nobody

knows how to construct one. This *probabilistic method* is now one of the most powerful tools in combinatorics.

Order from chaos

Ramsey's theorem has many relatives, all expressing the same idea: large enough structures cannot avoid order. *Van der Waerden's theorem* says that if the whole numbers are coloured with finitely many colours, there are arbitrarily long arithmetic progressions of one colour. The *Erdős–Szekeres theorem* says that any sequence of $n^2 + 1$ distinct numbers contains an increasing or a decreasing subsequence of length $n + 1$. And Ramsey himself was interested in a question of logic: he proved his theorem as a lemma in a paper on deciding the truth of logical statements.

Try this next

1 **At least two.** Show that in any two-colouring of K_6 there are in fact at least *two* one-colour triangles. (*Hint:* count the "bichromatic angles": pairs of edges of different colours at a common vertex. A point with g gold and $5 - g$ silver edges is at the apex of $g(5 - g) \leq 6$ such angles. Each triangle that is not one-colour contains exactly two.)

2 **Three colours.** If the edges of K_{17} are coloured in three colours, show that there is a one-colour triangle. (*Hint:* one point has at least six edges of one colour.)

3 **Three or four.** Show that $R(3,4) \leq 10$: in any gold/silver colouring of K_{10} there is a gold triangle or a silver K_4. (*Hint:* a point has either at least 4 gold edges or at least 6 silver edges.) In fact $R(3,4) = 9$.

4 **Monotone subsequences.** Prove the Erdős–Szekeres theorem for $n = 2$: any sequence of 5 distinct numbers contains an increasing or decreasing subsequence of length 3. Find a sequence of 4 with neither.

Further reading

- **Miklós Bóna, *A Walk Through Combinatorics*** (World Scientific), the chapters on the pigeonhole principle and Ramsey theory.
- **Ronald Graham & Joel Spencer, "Ramsey theory"**, *Scientific American* (July 1990). A popular introduction by two leading researchers.
- **Noga Alon & Joel Spencer, *The Probabilistic Method*** (Wiley). The standard book on Erdős's method; the first chapter is accessible.
- **Paul Hoffman, *The Man Who Loved Only Numbers*** (Fourth Estate). A biography of Erdős.

The full guided solution is in Section 9.2:
https://beyondstep.keithmcnulty.org/ch09/ramsey.html